

Package #163

Apply a permanent, covalently bonded hydrophilic polymer (e.g., PEG brush) to the catheter lumen and external surface to create a passive, nonfouling interface that physically repels protein and cellular adhesion, thereby preventing clot buildup without active heparin.

Technology Maturity: ENGINEERING_DEFINITION

Transfer Posture: Sponsored Validation

Problem: Documented problem 'catheter occlusion' concerning tunneled hemodialysis catheter. User-submitted engineering problem (session input): "Why do tunneled hemodialysis catheters lose flow patency within weeks of insertion despite flushing protocols?". [fda_maude] 100 records retrieved live (1992-2032). Illustrative records: 'MAUDE report 1000: Injury [SAME AS ABOVE]'; 'MAUDE report 10000024: Injury [CAPSURE SP NOVUS LEAD]'. LIMITS STAMPED INLINE: voluntary/regulatory reports; counts are NOT incidence rates; causality UNVERIFIED (Art. XXI.5). [europepmc] 10 records retrieved live (2026-2026). Illustrative records: 'Commentary on: "clinical and haemodynamic evaluation of a novel physician modified inner branch iliac branch device for East Asians".'; 'Biosurfactants as Antibiofilm Agents for Medical Devices: Mechanisms, Evidence and Integration into Infection Prevention and Control.'. LIMITS STAMPED INLINE: literature context establishes the problem class, not event rates. Problem-existence basis: live records retrieved.

Mechanism: Heparin coating provides an anticoagulant surface that inhibits thrombus formation, leading to longer failure-free survival.

Contents:

- 01 - Executive Technology Brief
- 02 - Engineering Technology-Transfer Dossier
- 03 - Buyer Decision Card
- 04 - Evidence Summary
- 05 - Transfer Manifest

Status: ENGINEERING_DEFINITION. Not physically validated.